CLINICAL · NECK, THYROID & SALIVARY

Cysts and tumours of the jaws

Most jaw lesions are found by chance on a dental radiograph, and the radiograph does most of the diagnostic work. Is the lesion radiolucent or radio-opaque? Single or multiple? Well defined or ragged? Unilocular or multilocular? Related to a tooth crown, a root apex, or neither? A periapical cyst on a dead tooth needs a root canal treatment. A keratocyst in a young adult raises the question of Gorlin syndrome. A multilocular "soap-bubble" lesion at the angle is an ameloblastoma until biopsy says otherwise.

AT A GLANCE

- Radicular cyst: the only cyst with a dead tooth.
- Dentigerous cyst: around an unerupted crown.
- Multiple keratocysts: Gorlin syndrome.
- Ameloblastoma: soap bubble at the angle; resect with margins.

1 CHAPTER ONE Approach

Reading the jaw.

1.1 Clinical assessment

Most lesions present as a painless bony expansion of the jaw. When the cortex has thinned it gives a characteristic "egg-shell crackling" on pressure, and the teeth over it may be displaced, loose or, if they never erupted, missing. Pain and infection usually mean that a benign lesion has become secondarily infected, but *numbness of the lower lip* from involvement of the inferior alveolar nerve points towards malignancy or osteomyelitis and should never be dismissed.

Test the vitality of the related teeth with cold or an electric pulp tester. The rule is simple and useful: *every jaw cyst except the periapical (radicular) cyst is associated with vital teeth.*

Aspirate before any incisional biopsy. Blood suggests a vascular lesion such as an arteriovenous malformation or aneurysmal bone cyst, and *a pulsatile or blood-filled*

lesion must never be biopsied blind, because torrential bleeding may follow. Straw-coloured fluid suggests a dentigerous or radicular cyst, cheesy keratin an odontogenic keratocyst, and a dry tap a solid tumour or a traumatic bone cavity.

1.2 Imaging

A panoramic radiograph (OPG) with intraoral films is the first step. CT or cone-beam CT then defines size, any cortical breach, the relation to the inferior alveolar canal and the maxillary sinus, and features of malignancy, while MRI shows soft-tissue extension.

RADIOLOGICAL GROUPS

PATTERN	EXAMPLES
Radiolucent, unilocular	Radicular, residual, dentigerous, lateral periodontal and nasopalatine cysts; odontogenic keratocyst; Stafne defect; unicystic ameloblastoma; adenomatoid odontogenic tumour
Radiolucent, multilocular ("soap bubble")	Ameloblastoma, odontogenic keratocyst, odontogenic myxoma, central giant cell granuloma, aneurysmal bone cyst, glandular odontogenic cyst, Pindborg tumour
Radiolucent, multiple	Keratocysts in Gorlin syndrome, Langerhans cell histiocytosis, cherubism, myeloma, metastases
Radiolucent, ill-defined	Osteomyelitis, osteoradionecrosis, MRONJ, malignancy, metastases, Langerhans cell histiocytosis ("floating teeth")
Radio-opaque, well defined	Odontoma, cementoblastoma, osteoma, focal cemento-osseous dysplasia
Radio-opaque, ill-defined	Fibrous dysplasia ("ground glass"), osteosarcoma ("sunburst"), chondrosarcoma, Paget disease ("cotton wool")
Mixed	Ossifying fibroma, calcifying odontogenic cyst, adenomatoid tumour, cemento-osseous dysplasia

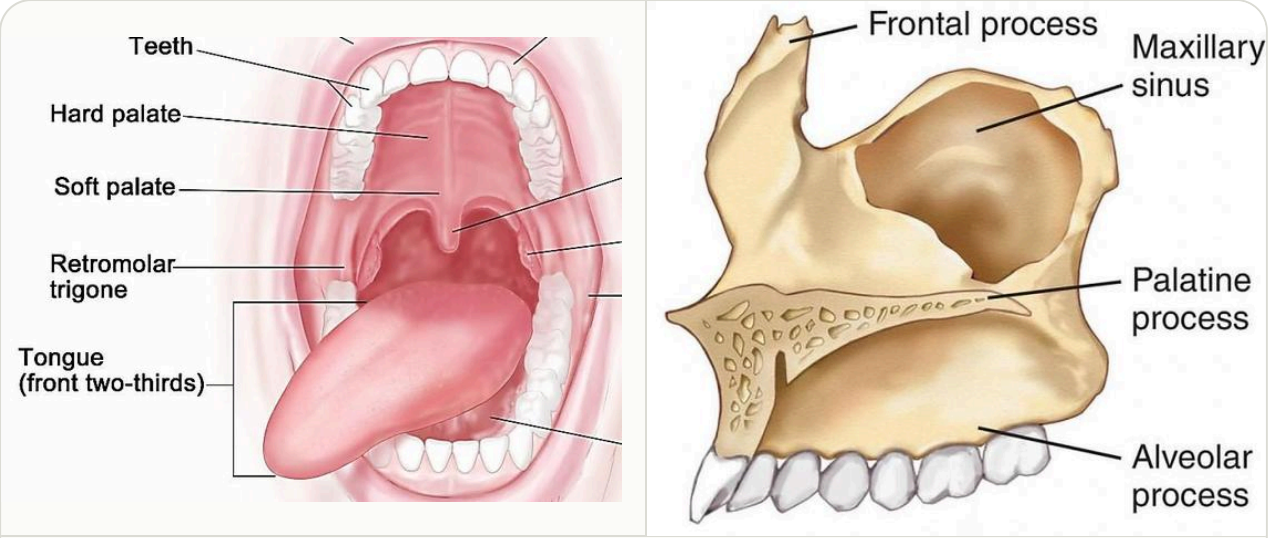


FIGURE 1 The teeth, palate and retromolar trigone, and the maxilla with its alveolar process.

Anatomy atlas · ENT Steps p. 172, ENT Steps p. 129

2 CHAPTER TWO Cysts of the jaws

Odontogenic and non-odontogenic.

A true cyst has an epithelial lining. Odontogenic cysts arise from the remnants of tooth-forming epithelium, and each remnant gives its own cyst: the *rests of Malassez* give radicular cysts, the *reduced enamel epithelium* gives dentigerous cysts, and the *dental lamina (rests of Serres)* gives keratocysts and lateral periodontal cysts.

2.1 Radicular (periapical) cyst

The radicular cyst is the *commonest jaw cyst*. It follows death of the pulp of a tooth, which leads to a periapical granuloma that then becomes cystic. On the radiograph it is a well-defined, round lucency *at the root apex* of the non-vital tooth, and it may announce itself with a sinus onto the gum or skin. It is treated by root canal treatment, or by extraction with enucleation of the cyst. A cyst that persists after the tooth has gone is a **residual cyst**, and is enucleated.

2.2 Dentigerous (follicular) cyst

The dentigerous cyst is the second commonest. It forms around the *crown of an unerupted tooth*, attached at the amelocemental junction, usually a lower third molar

or an upper canine, in patients aged 10 to 30. It expands painlessly and may push the tooth a long way, into the ramus or the antrum.

Most are enucleated with the tooth. Large cysts in children are better *marsupialised* or decompressed, which lets the cyst shrink and the tooth erupt. Rarely an ameloblastoma or carcinoma arises in the wall, which is why the lining is always sent for histology. Its soft-tissue counterpart in a child, the **eruption cyst**, is a bluish swelling of the gum over an erupting tooth that resolves as the tooth comes through.

2.3 Odontogenic keratocyst (OKC)

The keratocyst was renamed a "keratocystic odontogenic tumour" in 2005 and reclassified as a cyst in 2017, but whatever its label it behaves aggressively. It usually occupies the *posterior body and ramus of the mandible* of a young adult, and because it extends along the medulla with little expansion it is often large by the time it is found.

Histology shows a thin, *parakeratinised* lining with a palisaded, hyperchromatic basal layer and *daughter cysts* in the wall, and the aspirate is cheesy keratin. The friable lining and satellite cysts explain recurrence rates of up to 30%. Large cysts are first decompressed, then enucleated with a *peripheral ostectomy* or Carnoy solution; multiple recurrences need resection, and follow-up is long.

Multiple keratocysts, especially in a child, mean **Gorlin syndrome** (naevoid basal cell carcinoma syndrome), an autosomal dominant condition caused by a *PTCH1* mutation. Look for multiple basal cell carcinomas, palmar and plantar pits, a calcified falx, bifid ribs, frontal bossing with hypertelorism, and medulloblastoma, and avoid radiation in these patients. The **orthokeratinised odontogenic cyst** looks similar but has an orthokeratinised lining without palisading, rarely recurs and carries no syndrome.

2.4 Other odontogenic cysts

The **lateral periodontal cyst** is a small lucency between the roots of vital lower premolars and canines in adults. The **glandular odontogenic cyst** of the anterior mandible can be confused with central mucoepidermoid carcinoma, recurs, and may

need resection. The **calcifying odontogenic (Gorlin) cyst** contains "ghost cells" with calcification, and the **paradental (buccal bifurcation) cyst** lies buccal to an erupting lower molar in a child.

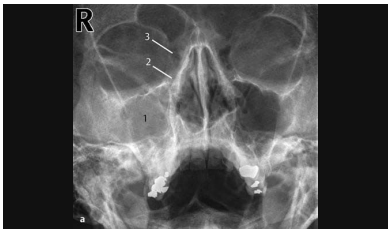
2.5 Non-odontogenic cysts

The **nasopalatine (incisive canal) cyst** is the commonest non-odontogenic cyst. It appears as a *heart-shaped* lucency between vital upper central incisors, with a swelling behind them, and is enucleated. The **nasolabial (nasoalveolar) cyst** lies in soft tissue beneath the ala, filling the nasolabial fold and lifting the nasal floor, mostly in women; it is lined by respiratory epithelium and is removed through a sublabial approach or marsupialised endoscopically into the nose.

Three lesions are pseudocysts, without an epithelial lining. The **traumatic (simple) bone cavity** is an empty space in a teenager that scallops between the roots and heals after surgical exploration. The **aneurysmal bone cyst** is a blood-filled expansile "soap bubble" treated by curettage. The **Stafne defect** is a lingual depression below the inferior alveolar canal at the angle, filled by submandibular gland, and needs no treatment.

IMAGING IN THE APP 1 case

Odontogenic disease extending into the maxillary sinus.



Odontogenic Sinusitis

Infection & inflammation · 3 images

3 CHAPTER THREE Odontogenic tumours

Most are benign.

3.1 Ameloblastoma

The ameloblastoma is the commonest clinically significant odontogenic tumour: benign, but *locally aggressive*. It affects adults in the third to fifth decades, and about 80% arise in the mandible, mostly in the *molar–ramus region*. It grows slowly and painlessly, displacing and loosening teeth and resorbing their roots.

The radiograph shows a *multilocular "soap-bubble"* or honeycomb lucency with cortical expansion and *knife-edge root resorption*. Histologically there are islands of odontogenic epithelium whose peripheral columnar cells palisade with *reverse polarisation* of their nuclei away from the basement membrane, in follicular, plexiform, acanthomatous or desmoplastic patterns. Most mandibular tumours carry *BRAF V600E*, and BRAF-targeted therapy is now used for unresectable or recurrent disease.

Several variants matter. The **unicystic** ameloblastoma of young patients often mimics a dentigerous cyst; its luminal type can be enucleated with close follow-up, but the mural type behaves like a solid tumour. The **peripheral** form is a gum mass. The **metastasising ameloblastoma** has benign histology yet spreads to lung or nodes, and **ameloblastic carcinoma** is frankly malignant.

Treatment is *resection with about 1 to 1.5 cm margins beyond the radiological edge*, including a soft-tissue margin wherever the cortex has been breached: a marginal or segmental mandibulectomy, or a partial maxillectomy, with reconstruction by a fibula free flap and dental implants. Enucleation and curettage give recurrence rates of 50 to 90%. Maxillary tumours are the more dangerous because they can reach the skull base, and follow-up is lifelong because recurrence may appear after decades.

OPERATIVE GUIDES IN THEATRE 3 operations

Marginal and segmental mandibulectomy, and microvascular reconstruction.

Marginal Mandibulectomy
Marginal mandibulectomy
Segmental Mandibulectomy
Segmental mandibulectomy
Microvascular Reconstruction of the Head and Neck
Free flap reconstruction

3.2 Other odontogenic tumours

The **odontoma** is the commonest odontogenic tumour overall, though really a hamartoma. *Compound* odontomas in the anterior maxilla contain small tooth-like structures, and *complex* odontomas in the posterior mandible are a disorganised mass of enamel and dentine. Both are radio-opaque with a lucent rim, often stop a tooth from erupting, and are enucleated.

The **adenomatoid odontogenic tumour** occurs in teenage girls in the anterior maxilla around an unerupted canine, as a lucency with fine calcifications, and is enucleated. The **calcifying epithelial odontogenic (Pindborg) tumour** of adults in the posterior mandible contains amyloid with concentric *Liesegang ring* calcifications and is resected with a margin. The **ameloblastic fibroma** of young patients is enucleated. The **odontogenic myxoma**, with its "soap bubble" or "tennis racket" lucency of stellate cells in myxoid stroma, is unencapsulated and infiltrative, and recurs in about a quarter even after resection. The **cementoblastoma** is a radio-opaque mass fused to a lower molar root with a lucent halo, removed with the tooth.

4 CHAPTER FOUR Non-odontogenic lesions

Bone, fibro-osseous, systemic.

4.1 Fibro-osseous lesions

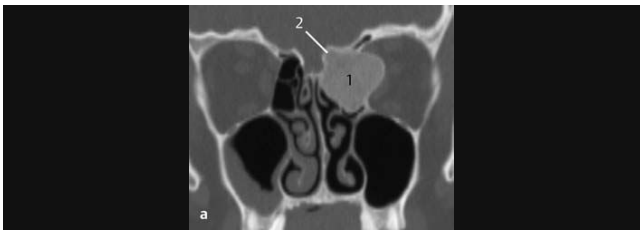
In **fibrous dysplasia** bone is replaced by fibrous tissue and woven bone as a result of a *GNAS* mutation. About 80% is monostotic; polyostotic disease with café-au-lait macules and endocrine overactivity such as precocious puberty is *McCune–Albright syndrome*. It causes painless facial asymmetry in a child or teenager, the

maxilla more often than the mandible, and growth usually slows after puberty. The radiograph shows an ill-defined *ground-glass* expansion blending into normal bone. Most patients are observed, recontouring is done for deformity once growth stops, and the optic canal is decompressed if vision is threatened.

The **ossifying fibroma**, by contrast, is a true, well-demarcated neoplasm of mixed lucency and opacity, typically in the mandible of young women; its juvenile trabecular and psammomatoid forms are aggressive in children and in the sinuses. It is enucleated or resected. **Cemento-osseous dysplasia**, periapical, focal or florid, occurs in middle-aged women of African or Asian descent and is best left alone, because extraction is followed by infection.

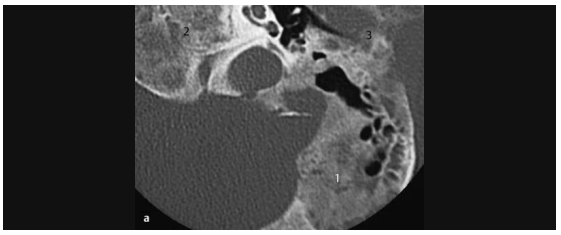
IMAGING IN THE APP 2 cases

Fibrous dysplasia on imaging.



Fibrous Dysplasia of the Sinuses (Monostotic and Polyostotic)

Bone disease · 5 images



Fibrous Dysplasia of the Temporal Bone and Skull Base

Bone disease · 7 images

4.2 Giant cell lesions

The **central giant cell granuloma** affects patients under 30 in the anterior mandible and may cross the midline. Its multinucleated giant cells are indistinguishable from those of a *brown tumour of hyperparathyroidism*, so calcium and PTH must always be checked. It is treated by curettage or with intralesional steroids, calcitonin or denosumab.

Cherubism is autosomal dominant and produces painless *bilateral* swelling of the mandibular angles, and often the maxillae, from the age of two to five, giving the

"cherub" face with upturned eyes. The radiograph shows multilocular lucencies, and the condition regresses after puberty.

4.3 Bone tumours

An **osteoma** is a dense, well-defined mass. Multiple osteomas with colonic polyposis and soft-tissue tumours make up *Gardner syndrome* (an APC mutation), and colonoscopy is mandatory because the polyps become malignant. The **osteoblastoma** is painful, and any cartilaginous jaw tumour, even one called a chondroma, should be treated as a possible chondrosarcoma.

Osteosarcoma of the jaw occurs a decade or two later than in the limbs, in the third and fourth decades, and also follows radiotherapy and Paget disease. It presents with swelling, pain, loose teeth and lip numbness, and the radiograph shows a *sunburst* periosteal reaction and *symmetrical widening of the periodontal ligament space* (Garrington sign). It needs radical resection with clear margins and chemotherapy. **Chondrosarcoma** affects older adults, the anterior maxilla more than the mandible, and needs radical resection because it resists radiotherapy and chemotherapy.

Metastases from breast, lung, prostate, kidney and thyroid usually land in the posterior mandible and present with pain, lip numbness or an extraction socket that will not heal; the "*numb chin*" can be the first sign of cancer. Myeloma and lymphoma give punched-out lesions, and **Langerhans cell histiocytosis** in children produces punched-out lucencies around the roots so that the teeth seem to float ("*floating teeth*"); it is treated by curettage, steroids, low-dose radiotherapy or chemotherapy. Finally, **squamous carcinoma** of the overlying mucosa invades the mandible through the occlusal ridge or the tooth sockets.

4.4 Paget disease

Paget disease is disordered bone remodelling in older people. The maxilla and skull enlarge with "*cotton-wool*" radiodensities, the roots show hypercementosis, the alkaline phosphatase rises, and bony overgrowth can compress cranial nerves and cause hearing loss. It is treated with bisphosphonates.

5 CHAPTER FIVE Osteonecrosis and infection

Dead bone in the jaw.

5.1 Osteomyelitis

Osteomyelitis usually affects the mandible after dental infection or a fracture, and diabetes and immunosuppression predispose to it. It causes pain, swelling, pus and sequestra, and numbness of the lower lip (*Vincent sign*). Treatment is debridement, sequestrectomy and prolonged antibiotics.

5.2 Osteoradionecrosis

Osteoradionecrosis follows radiotherapy, especially above 60 Gy. It is prevented by dental treatment before radiotherapy, and treated along a ladder from debridement and pentoxifylline–tocopherol to resection with a free fibula flap.

5.3 Medication-related osteonecrosis (MRONJ)

MRONJ is exposed bone for more than eight weeks in a patient taking antiresorptive drugs such as bisphosphonates or denosumab, or antiangiogenic drugs, who has never had radiotherapy to the jaw. The risk is highest with intravenous treatment for cancer, and it usually follows an extraction, so dental treatment should be completed before the drugs are started. It is managed with chlorhexidine, antibiotics and conservative debridement, with resection for advanced disease.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Systematic reading of a jaw radiograph,** and why aspiration comes before biopsy.
- **Radicular versus dentigerous cyst:** the tooth relationship and vitality.
- **Odontogenic keratocyst:** its lining, why it recurs, and Gorlin syndrome.
- **Ameloblastoma:** age, site, soap-bubble picture, reverse polarity, resection margins, BRAF.
- **Fibrous dysplasia** (ground glass) versus ossifying fibroma (well-defined).
- **Giant cell granuloma:** exclude hyperparathyroidism.
- **Osteosarcoma signs;** the numb chin.
- **Osteoradionecrosis and MRONJ.**

6.2 Common pitfalls

COMMON PITFALLS

- Biopsying a jaw lucency without aspirating first, and meeting an AVM.
- Curetting an ameloblastoma.
- Missing Gorlin syndrome in a young patient with several keratocysts.
- Calling a central giant cell lesion without checking calcium and PTH.
- Extracting teeth in an irradiated jaw or in a patient on intravenous bisphosphonates without precautions.

Written for this app from K. J. Lee ch. 39. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

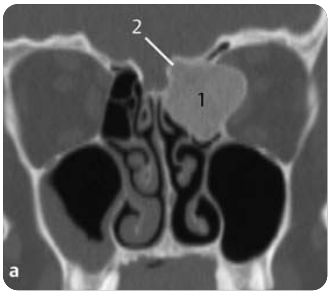
On imaging

The imaging of this topic: key findings, differentials and representative scans.

PARANASAL SINUSES · BONE DISEASE

Fibrous Dysplasia of the Sinuses (Monostotic and Polyostotic)

-
- Expansion of bone with a ground glass appearance and intact cortex reflects the pathophysiology of this disease (i.e., replacement of normal medullary bone by woven fibro-osseous tissue).
- The lesion may be misinterpreted as a mucocele or an osteoma, depending on the CT window setting.
- An ossifying fibroma may demonstrate central bony densities in later stages.
- Secondary sinusitis and vascular or neurological deficits may develop with progression of the lesions due to encroachment of sinus ostia, fissures, skull-base foramina, and/or the optic canal. Esthetic and functional problems due to lesions in the mandible and maxilla are not uncommon.
- Expansion of bone with a ground glass appearance and intact cortex reflects the pathophysiology of this disease (i.e., replacement of normal medullary bone by woven fibro-osseous tissue). –
- A separate variant of polyostotic fibrous dysplasia is seen in McCune–Albright syndrome, with severe progressive lesions in multiple locations (also seen more often in females, and with cutaneous pigmentations and precocious puberty).

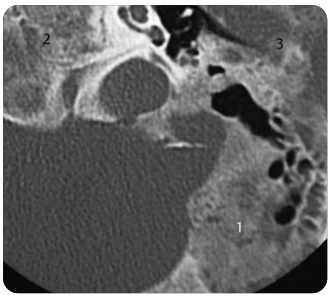


CT, coronal, bone-tissue setting. A solitary lesion (1) is located in the left ethmoid and is obstructing the frontal recess with expansion into the left…

TEMPORAL BONE & SKULL BASE · BONE DISEASE

Fibrous Dysplasia of the Temporal Bone and Skull Base

- Radiologically on CT, enlargement of bone with intact cortical outlines and a dense ground glass appearance is characteristic of fibrous dysplasia. Fibrous dysplasia incidentally found on MRI may be confused with a malignancy.
- In meningioma, bony outlines can be irregular and signs of hyperostosis can be found as well as a dural tail. Sometimes intracranial calcifications might be indicative of an intracranial meningioma.
- In progressive disease, there is encroachment of neural structures and foramina in the skull base, with resultant neurologic and vascular deficits. The external auditory canal may be narrowed and medial to the stenosis; an inclusion cholesteatoma might develop.
- Surgical procedures are limited to prevent complications of encroachment of critical structures and treatment of chronic infections in the middle ear cavity. In obstruction of the auditory canal, extra attention must be paid to the potential risk of inclusion cholesteatoma.
- Fibrous dysplasia in a single osseous structure (monostotic) can be monitored with serial scanning and can be stable and asymptomatic for years.
- In cases of meningioma, the bony outlines can be irregular and hyperostosis can be found as well as a dural tail. Intracranial calcifications may be suggestive of a meningioma.
- In later stages of Paget disease, the otic capsule is completely resorbed. Osteopetrosis demonstrates a complete lack of pneumatization of the temporal bone. Craniometaphyseal dysplasia is recognized by the presence of increased bone densities.
- Beware of encroachment of vital structures and cranial nerves as well as the formation of an inclusion cholesteatoma.

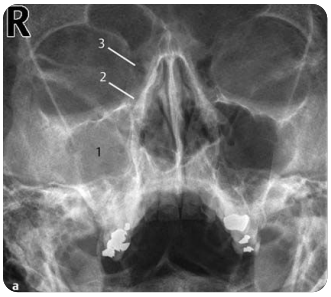


CT, axial. Fibrous dysplasia is characterized by replacement of medullary bone by abnormally proliferating fibrous tissue, resulting i…

PARANASAL SINUSES · INFECTION & INFLAMMATION

Odontogenic Sinusitis

- Destruction of bony outlines with infiltrative characteristics is suggestive of inverted papilloma, malignancies, or fungal infections.
- Calcifications in the opacified area are suggestive of fungal infections.
- History of frequent dental consultations and resorption of bone around tooth roots may indicate odontogenic infections.
- Long-term antibiotics are an essential part of postoperative treatment.



Plain radiograph, Waters view. There is opacification of the right maxillary sinus (1) as well as in the region of the right ethmoid (2). The right frontal…